

LIGHTWEIGHT SAFE HELICOPTER (LSH)

DOCUMENT 00 — COMPREHENSIVE ENGINEERING
BASELINE

Generation 1 — LSH V4.0

20-TON-CLASS INTEGRATED SURVIVAL HELICOPTER

Separable Occupied Survival Capsule + Controlled Rapid Separation +
Multi-Mode Recovery Architecture

Document Title	LSH V4.0 — Comprehensive Engineering Baseline
Document Number	LSH-00-400
Document Type	Master Engineering Specification — Conceptual / IP Baseline
Document Class	Conceptual Engineering / Intellectual Property Development Record
Revision	V4.0 — Generation 1, 20-ton-class baseline
Status	[PROPOSED] Preliminary Controlled Engineering Baseline
Development Stage	Pre-Simulation / Pre-Hardware / Pre-Prototype
Language	English
Confidentiality	Proprietary Concept Development Material
Founder	Amirbahador Atai

Engineering integrity statement: This document defines a controlled conceptual baseline. Numerical values marked **TARGET, ASSUMPTION, STUDY, CALC** or **OPEN** are not demonstrated performance, certification results, or flight-proven values. Critical claims require analytical closure, simulation, ground testing, integrated testing and, where applicable, flight-test evidence.

Revision intent: supersede the previous Generation 1 planning baseline of 15,000 kg / 15 m main rotor / 2.7 m tail rotor / 750–900 km range with a new controlled LSH V4.0 study baseline of 20,000 kg / 21 m main rotor / 3.5 m tail rotor / 1,000 km range target, while preserving and strengthening the survival-capsule, rapid-separation and independent-safety architecture.

1. PURPOSE AND SCOPE

LSH V4.0 Generation 1 defines the system-level architecture for a 20-ton-class single-main-rotor helicopter whose primary distinguishing engineering concept is an occupied survival capsule that can become the final landing boundary during a catastrophic emergency. The aircraft is designed around a layered survival chain rather than reliance on a single emergency device.

The V4.0 baseline covers the aircraft, occupied survival capsule, rapid selective separation system, safety initiation paths, deterministic safety authorization, stabilization, recovery systems, impact attenuation, post-event survival and the interfaces required to verify the complete chain.

The design philosophy is not 'separate everything whenever the aircraft starts descending'. Separation is a controlled emergency mode authorized only when defined conditions are satisfied. The system must account for altitude, airspeed, attitude, angular rates, structural state, fire state, rotor state, available recovery modes, capsule clearance and downstream hazards.

The survival architecture is intended to support multiple emergency outcomes: conventional flight recovery where possible, autorotation where applicable, selective hazard/mass reduction, capsule separation, parachute-assisted descent, energy-absorbing landing and post-landing rescue. The architecture must not require the most extreme survival mode for every failure.

The baseline also preserves human emergency authority. Automatic systems may initiate or recommend a survival sequence only within defined authorization envelopes; the crew must retain an independent means to command, inhibit or terminate automatic action where technically and physically possible.

This document is an engineering/IP baseline, not a certification approval. Airworthiness, structural substantiation, parachute qualification, release reliability, occupant injury criteria, flight performance and global patent novelty remain open work items.

2. BASELINE STATUS AND CONFIGURATION CONTROL

Status Label	Meaning in this document
[LOCKED CONCEPT]	Architecture intentionally retained as a baseline design principle; implementation remains subject to engineering verification.
[REQ]	Top-level requirement proposed for subsequent requirement allocation.
[TARGET]	Numerical or functional development objective; not demonstrated.
[ASSUMPTION / STUDY]	Working engineering assumption used for preliminary sizing/accounting.
[CALC]	Value calculated from stated assumptions and equations.
[OPEN]	Requires analysis, simulation, test, supplier data or design decision.
[TBD — DEPENDENCY]	Cannot be responsibly fixed until a dependent design activity closes.
[PRIOR ART STATUS: NOT YET SEARCHED]	No novelty conclusion is implied.

The V4.0 baseline shall be configuration-controlled. Any change to MTOW, rotor diameter, capsule mass, occupant count, separation sequence, parachute architecture, landing-support geometry or safety-layer independence shall trigger an impact assessment across mass properties, CG, inertia, dynamics, structures, controls, recovery and verification.

3. GENERATION 1 V4.0 TOP-LEVEL TECHNICAL BASELINE

Parameter	LSH V4.0 Gen1	Status
Maximum Take-Off / Design Mass	20,000 kg	[TARGET]
Occupancy	10–12 persons	[TARGET / STUDY]
Flight architecture	Conventional single main rotor + tail rotor	[REQ / STUDY]
Main rotor diameter	21.0 m	[TARGET]
Main rotor disk area	346.36 m ²	[CALC]
Weight-based disk loading	57.7 kg/m ² (566 N/m ²)	[CALC]
Tail rotor diameter	3.5 m	[TARGET]
Tail rotor disk area	9.62 m ²	[CALC]
Propulsion architecture	Two turboshaft-class engines	[REQ / STUDY]
Landing gear	No retractable wheeled gear; 8 short, thick fixed mechanical supports integrated with capsule	[TARGET / CONCEPT]
Range	1,000 km	[TARGET]
Hover endurance	To be derived after power/fuel/thermal closure	[OPEN]
Survival capsule	Dedicated occupied final landing boundary	[REQ]
Capsule preliminary complete mass	5,200 kg	[TARGET]
Capsule recovery	Distributed multi-canopy system; 8-canopy architecture for preliminary study	[STUDY]
Airframe recovery parachute	Optional independent residual-airframe trade study	[OPTION / OPEN]
Rapid separation	Selective separation of designated high-mass/fire-hazard components	[REQ / CONCEPT]
Separation development objective	≤ 6 s from validated command to verified capsule-clear state	[TARGET / OPEN]

The 21 m rotor is intentionally larger than the previous 15 m planning rotor. At 20,000 kg, the resulting weight-based disk loading is approximately 57.7 kg/m², calculated from rotor disk area. This is a sizing indicator, not a guarantee of hover, climb, OEI or hot/high performance.

Preliminary ideal hover power is approximately 2.98 MW at sea-level standard density. Applying an illustrative figure-of-merit assumption of 0.75 gives approximately 3.98 MW rotor power. This is a study calculation only. Installed engine power must be determined from a complete power-required/power-available model including profile power, induced power, transmission losses, control margins, temperature, density altitude, rotor speed, transients and OEI requirements.

The 1,000 km range is a design target. It shall not be presented as achieved range until the mission model closes with aircraft drag, rotor performance, engine specific fuel consumption, fuel capacity, reserves, payload, temperature/altitude and operational constraints.

4. AIRCRAFT GEOMETRY AND CONFIGURATION

The V4.0 geometry below is a preliminary architecture sized around the 20-ton class. Dimensions other than rotor diameters are deliberately kept as ranges or open items until aerodynamic, structural, cabin, capsule and systems layouts are reconciled.

Geometry Item	V4.0 Preliminary Definition	Status
Overall length	Approx. 16.5–18.0 m	[TARGET / STUDY]
Maximum fuselage width	Approx. 4.0–4.5 m	[TARGET / STUDY]
Overall height	Approx. 4.8–5.3 m	[TARGET / STUDY]
Main rotor diameter	21.0 m	[TARGET]
Tail rotor diameter	3.5 m	[TARGET]
Rotor arrangement	Single main rotor + tail rotor	[REQ / STUDY]
Landing supports	8 fixed short mechanical supports under capsule	[TARGET]
Retractable landing gear	None	[LOCKED CONCEPT]
Capsule position	Central occupied boundary integrated within primary airframe	[LOCKED CONCEPT]
Capsule normal-flight load path	Dedicated primary attachment system with controlled emergency release	[REQ]
Capsule emergency load path	Capsule becomes independent final landing element after verified clearance	[REQ]

The eight landing supports shall not be treated as simple struts attached to a thin capsule shell. The intended load path is: ground contact → foot/ground interface → progressive energy absorber → lower capsule load path → primary capsule structure → occupant restraint/seat energy management. Structural sizing shall consider vertical, longitudinal, lateral and torsional loads, slope, uneven terrain, soft soil, asymmetric touchdown and failed-leg cases.

Fixed short supports eliminate the mechanical complexity and failure modes associated with retractable landing gear, but their permanent external geometry must be included in aerodynamic drag, ground clearance, rotor downwash, structural loads and emergency separation analyses.

5. PRELIMINARY MASS ACCOUNTING — 20,000 KG BASELINE

The following accounting model is intentionally a planning model. It is constructed to close to 20,000 kg and must not be interpreted as a verified mass properties report.

Mass Group	Planning Mass	Status
Survival capsule, complete	5,200 kg	TARGET
Main airframe excluding capsule	4,300 kg	ASSUMPTION / STUDY
Main rotor, hub and controls	1,700 kg	ASSUMPTION / STUDY
Tail rotor, gearbox and drive	380 kg	ASSUMPTION / STUDY
Engines, transmission, gearboxes and mounts	2,300 kg	ASSUMPTION / STUDY
Fuel system + usable fuel allocation	2,700 kg	ASSUMPTION / STUDY
Rapid separation system + independent hardware	850 kg	ASSUMPTION / STUDY
Avionics, electrical, hydraulic, thermal and control equipment	1,350 kg	ASSUMPTION / STUDY
Growth / configuration reserve	1,220 kg	PROGRAM RESERVE
TOTAL	20,000 kg	[CALC / ACCOUNTING CLOSURE]

The complete capsule allocation of 5,200 kg is a preliminary target and includes the survival structure, seats/restraints, eight landing supports and energy absorbers, recovery parachute system, separation interfaces, internal emergency systems and essential post-landing survival provisions. The detailed capsule mass shall later be decomposed bottom-up.

The 2,700 kg fuel-system/usable-fuel allocation is a study allocation only. It must be reconciled against actual fuel density, tank geometry, usable volume, unusable fuel, system mass, reserve policy and the 1,000 km mission model. No specific fuel volume is locked by this document.

The 850 kg rapid-separation allocation is also a study value. It includes release mechanisms, local structural reinforcement, independent actuation power/signalling, locking/inhibit hardware, sensors and associated wiring/plumbing. It shall be refined after the actual jettison architecture is defined.

Mass properties shall be maintained for: normal MTOW, minimum mission mass, maximum fuel, minimum fuel, forward/aft CG, lateral CG, capsule-attached configuration, partial separation, asymmetric separation, failed release, capsule-only descent and post-landing configuration.

6. PROPULSION, ROTOR AND PERFORMANCE CLOSURE

6.1 Propulsion architecture

- Two turboshaft-class engines are retained as the Generation 1 baseline architecture.
- No engine model or exact shaft power is locked in Document 00 V4.0.
- Engine selection shall be based on total installed power, OEI capability, hot/high operation, transient response, fuel consumption, thermal margins, transmission compatibility, maintenance and availability.
- The engine installation shall support controlled emergency isolation and, where technically justified, separation from the survival capsule.
- Fuel isolation must be coordinated with emergency separation and fire containment; separation shall not create a new uncontrolled fuel hazard.

6.2 Main rotor

- Nominal main rotor diameter: 21.0 m [TARGET].
- Disk area: 346.36 m² [CALC].
- Weight-based disk loading at 20,000 kg: 57.7 kg/m² [CALC].
- Rotor blade mass, chord, twist, solidity, tip speed, rotor RPM, airfoil family and control authority remain [OPEN].
- The rotor shall be analyzed for hover, climb, cruise, maneuver, autorotation, transient and emergency states.
- Any emergency rotor release concept is conditional. The main rotor shall not be designated universally jettisonable until structural, dynamic, aerodynamic, gyroscopic, trajectory, clearance and post-release control analyses demonstrate acceptability.

6.3 Tail rotor

- Nominal diameter: 3.5 m [TARGET].
- Disk area: 9.62 m² [CALC].
- Final tail rotor size shall be closed against main rotor torque, tail-arm geometry, anti-torque authority, directional control margins, OEI states, crosswind and emergency configurations.
- Tail rotor, gearbox and drive may be candidates for controlled separation if the post-release dynamics are acceptable.

6.4 Performance closure requirements

- Power required versus power available over gross mass, altitude, temperature and rotor RPM.
- Hover ceiling and hot/high hover.
- Rate of climb and cruise performance.
- One-engine-inoperative performance and safe operating envelope.
- Autorotation performance and rotor energy management.
- Transmission and gearbox thermal/torque limits.
- Rotor vibration and dynamic stability.
- Cruise drag and fuel flow.
- Mission fuel including reserves.
- 1,000 km range closure.

7. 1,000 KM RANGE TARGET — MISSION DEFINITION

The 1,000 km figure is retained as a high-value product requirement target, but the document deliberately does not fabricate a final cruise speed, fuel burn or reserve figure. These values depend on the selected rotor, airframe drag, engine, fuel system and mission profile.

Mission Variable	V4.0 Baseline	Closure Requirement
Design range	1,000 km	[TARGET]
Cruise speed	TBD by aerodynamic/propulsion study	[OPEN]
Cruise altitude	TBD by mission study	[OPEN]
Fuel quantity	TBD by tank + mission closure	[OPEN]
Reserve policy	TBD by operational/certification basis	[OPEN]
Payload/occupancy	10–12 occupants + mission equipment	[TARGET]
Hot/high mission	Required	[REQ]
OEI diversion/reserve case	Required for safety assessment	[REQ]

The range model shall explicitly account for start/taxi or equivalent ground operation, climb, cruise, descent, loiter/diversion where applicable, reserve, unusable fuel, engine/transmission margins and environmental conditions. Range shall not be derived from fuel quantity alone.

8. OCCUPIED SURVIVAL CAPSULE

The occupied survival capsule is the central safety boundary of LSH V4.0. During normal flight it is structurally integrated into the aircraft through controlled primary attachments. During a validated catastrophic survival sequence, it is intended to become an independent occupied landing element.

Capsule Parameter	V4.0 Definition	Status
Occupancy	10–12 persons	[TARGET / STUDY]
Complete capsule mass	5,200 kg planning target	[TARGET]
Primary structure	Continuous CFRP roll-cage with metallic/titanium nodes and controlled joints	[CONCEPT]
Seats	Energy-absorbing seats with five-point/appropriate restraints	[TARGET]
Landing supports	8 short, thick fixed mechanical supports with progressive energy absorption	[TARGET]
Parachute architecture	Distributed multi-canopy; 8-canopy preliminary study architecture	[STUDY]
Capsule emergency systems	Independent emergency power, sensing, locator, communications and essential life-support	[STUDY]
Capsule clearance	Verified before recovery deployment	[REQ]
Post-landing stability	Eight-point support architecture with asymmetric/failed-leg analysis	[REQ]

The capsule shall be designed as a real structural system rather than a passenger cabin merely surrounded by a shell. The engineering program shall address load paths, crash loads, fatigue, damage tolerance, joints, attachment fittings, seat loads, floor loads, fire/smoke, environmental exposure, water/ditching provisions where required, emergency egress and occupant injury criteria.

The capsule shall retain sufficient independent capability after separation to support recovery and rescue. Essential systems should not depend on the main airframe power bus after separation.

9. EIGHT FIXED SHORT MECHANICAL LANDING SUPPORTS

LSH V4.0 uses eight short, thick, fixed mechanical supports located below the survival capsule. The concept is intentionally different from conventional retractable landing gear: the supports are part of the survival capsule's impact-attenuation system and remain with the occupants after separation.

- Eight supports provide load distribution and redundancy against uneven terrain and partial failure.
- Each support shall include a defined mechanical energy-absorption function rather than relying solely on elastic deformation.
- Replaceable or inspectable sacrificial/crush elements should be considered to improve post-event maintainability.
- The attachment system shall transmit controlled loads into the capsule's primary lower structure.
- The system shall be analyzed for vertical, lateral, longitudinal and combined impact loads.
- Asymmetric landing with one or more failed/degraded supports is a required analysis case.
- Terrain cases shall include hard flat ground, sloped ground, uneven ground and soft/low-bearing-strength surfaces.
- Ground clearance and rotor-downwash interaction shall be evaluated.
- Post-landing stability, tipping resistance and rescue access shall be included.

Preliminary capsule impact energy follows $E = 1/2 m v^2$. For a 5,200 kg capsule, illustrative kinetic energies are approximately 93.6 kJ at 6 m/s, 166.4 kJ at 8 m/s and 260.0 kJ at 10 m/s. These are not allowable impact energies; they are sizing illustrations only. Actual survival depends on energy absorption, occupant pulse, attitude/rates, terrain and structural response.

10. SIX-LAYER SURVIVAL AND SAFETY ARCHITECTURE

The V4.0 architecture retains and strengthens the six-layer functional survival philosophy from the previous LSH baseline. The layers are functionally distinct. Duplication alone shall never be used as evidence of independence. Independence shall be assessed for power, signal routing, sensors, processors, physical location, structural vulnerability, fire, EMI/lightning, common-cause failures and human interaction.

Layer	Function	Primary Purpose
L1	Primary automatic emergency detection and initiation	Detect catastrophic states and initiate a validated survival sequence when authorization conditions are met.
L2	Independent backup initiation	Provide a second physically/electrically segregated automatic path if L1 is unavailable.
L3	Independent manual/crew authority	Allow crew initiation, inhibit and emergency authority without dependence on L1/L2.
L4	Rapid selective physical separation	Separate designated mass/fire hazards from the capsule using controlled release mechanisms.
L5	Direct-impact survival	Protect occupants through capsule structure, restraints, seats and eight energy-absorbing supports.
L6	Recovery and post-event survival	Deploy recovery system, stabilize landing, locate occupants and support post-landing survival/rescue.

The architecture shall be evaluated as a chain, but it shall also be designed so that failure of one layer does not automatically eliminate all survival capability. For example, loss of automatic detection must not eliminate manual authority; loss of one release path must not necessarily eliminate all separation capability; and imperfect separation must not make the capsule structurally incapable of surviving residual loads.

The design principle is: **AI or advanced prediction may assist detection, classification and decision support; deterministic safety logic authorizes safety-critical actuation.** Confidence or model output shall never be treated as equivalent to safety authorization.

11. DETERMINISTIC SAFETY AUTHORIZATION AND SAFETY KERNEL

The Safety Kernel is the deterministic authorization boundary between emergency detection/decision logic and physical survival actuation. Its purpose is to prevent a computationally complex or uncertain decision layer from directly commanding hazardous separation without satisfying explicit safety preconditions.

The Safety Kernel shall evaluate, as applicable:

- Altitude above terrain and minimum recovery altitude.
- Airspeed and dynamic pressure envelope.
- Pitch, roll, yaw and angular-rate limits.
- Rotor RPM, rotor state and rotor control state.
- Engine/fire/fuel hazard state.
- Structural health and confirmed structural failure state.
- Capsule attachment health and release readiness.
- Separation trajectory clearance.
- Potential collision between released components and capsule/parachutes.
- Parachute deployment envelope and recovery-system availability.
- Landing support availability and degraded configurations.
- Sensor validity and disagreement.
- Power availability for release and recovery systems.
- Communication availability where communication is part of the architecture.
- Manual crew authority and emergency override state.
- Rollback/fallback availability.

If the safety conditions are not satisfied, the system shall inhibit the unsafe action or transition to a defined degraded mode. The design shall define what constitutes 'no safe optimal solution': safety and physical feasibility take priority over optimization or mission continuation.

No software model, AI classifier, optimizer or predictive system shall bypass the Safety Kernel, aircraft protection systems, interlocks, equipment limits or human emergency authority.

12. CONTROLLED RAPID SEPARATION SYSTEM

12.1 Separation philosophy

The rapid separation system is a selective, staged, controlled mechanism for reducing the mass, fire load and rotating/energetic hardware that must remain associated with the occupied capsule during the final descent. It is not a blanket requirement to release every aircraft component in every emergency.

The system shall use state-dependent release logic. Components shall be designated as jettisonable only after structural, aerodynamic, dynamic, fire, control, trajectory and collision analysis demonstrates that release is acceptable within a defined envelope.

12.2 Candidate separable components

- Main engines and associated high-mass engine installation interfaces, where practical.
- Selected transmission/drive components where release is structurally and dynamically acceptable.
- Main rotor blades and/or hub components only if detailed dynamic and trajectory analysis demonstrates safe clearance; main rotor release is NOT universally locked.
- Tail rotor, tail-rotor gearbox and drive shafting where release is safe.
- Primary fuel tanks and major fuel lines that constitute a significant fire hazard, after safe fuel isolation.
- Selected external structural panels or high-mass attachments that do not form part of the capsule primary load path.
- Other components only through configuration-controlled hazard and trajectory analysis.

12.3 Mandatory release protections

- Positive mechanical retention during normal flight.
- Electrical/logic inhibits against inadvertent release.
- Independent command paths.
- Defined arm/fire sequence.
- Release-state sensing.
- Fuel and energy isolation before or coordinated with release.
- Asymmetric/partial-release detection.
- Post-release trajectory monitoring where practical.
- Capsule-clear verification before parachute authorization.
- Defined failed-release and rollback/fallback behavior.

13. PRELIMINARY SIX-SECOND SURVIVAL SEQUENCE

The following sequence is a development objective inherited and strengthened from the prior LSH rapid-jettison baseline. It is intentionally aggressive and shall remain [OPEN] until mechanism tests and dynamic simulations demonstrate that timing, reliability and separation quality can be achieved.

Time	Action	Safety Purpose
0.0–0.4 s	Validate emergency state; arm authorized release path; isolate fuel/electrical hazards	Prevent emergency hazards or an invalid or unsafe state.
0.4–2.2 s	Stage primary high-mass/rotating-system release where authorized	Reduce major mass and energy/fire hazards.
2.2–3.8 s	Release tail-rotor/drive and primary fuel hazards where authorized	Remove remaining major hazards.
3.8–5.4 s	Release remaining designated airframe/capsule attachments	Free the capsule from the main airframe.
5.4–6.0 s	Verify capsule clearance and recovery envelope	Prevent premature parachute deployment.
After clearance	Authorize capsule recovery system	Transition to capsule-only or capsule-dominant survival state.

The sequence shall not be treated as a fixed certification requirement. If the detailed architecture cannot safely achieve ≤ 6 s, the target shall be revised under configuration control rather than achieved by reducing verification rigor.

Partial and asymmetric separation are first-class emergency states. The system shall not assume that every release mechanism functions perfectly. It must recognize which components remain attached and determine whether the resulting state is survivable, whether another controlled action is permitted, or whether the capsule must proceed with degraded recovery.

14. MULTI-MODE RECOVERY ARCHITECTURE

The LSH V4.0 recovery architecture is intentionally layered so that survival is not dependent on one device.

Recovery Mode	Use Case	Status
Controlled flight recovery	Recoverable control/propulsion disturbance	[REQ / STUDY]
Autorotation	Engine/power loss when altitude/rotor state permit	[REQ / STUDY]
Selective mass/hazard separation	Fire, structural or combined emergency requiring reduction of carried hazard/loss	[REQ / CONCEPT]
Capsule separation	Catastrophic state where independent occupied recovery is preferable	[REQ / CONCEPT]
Distributed capsule parachutes	Primary post-separation vertical recovery	[STUDY — 8 CANOPIES]
Eight energy-absorbing supports	Final ground impact attenuation	[REQ / TARGET]
Optional airframe recovery parachute	Residual-airframe recovery or unmanned asset protection	[OPTION / TRADE STUDY]
Post-landing survival	Locator, communications, lighting, emergency power and rescue interface	[REQ]

The capsule parachute system should be distributed rather than treated as a single-point canopy where practical. The final number, canopy size, deployment sequence, reefing, riser architecture, failure tolerance and inflation dynamics require parachute sizing and test.

An independent parachute for the residual main airframe is retained as an optional trade study, not a mandatory V4.0 requirement. It can be valuable for reducing ground hazard, protecting unmanned equipment or providing an additional recovery mode, but it introduces additional mass, drag, deployment hazards and potential interaction with the capsule. Its value shall therefore be established through system-level trade analysis.

15. EMERGENCY STATE MACHINE

LSH V4.0 shall use an explicit emergency state machine rather than a single binary 'normal/emergency' flag.

- 1 NORMAL FLIGHT — all safety layers monitored; release hardware mechanically locked.
- 2 ANOMALY DETECTED — sensor/model detects a potential abnormal condition.
- 3 CONFIRMATION — independent evidence and validity checks determine whether the condition is real.
- 4 IMPACT ASSESSMENT — estimate control authority, altitude, airspeed, fire/structural risk, rotor/engine state and available recovery modes.
- 5 RECOVERY ATTEMPT — use conventional control, engine redundancy or autorotation where safe and appropriate.
- 6 SURVIVAL AUTHORIZATION — Safety Kernel evaluates whether selective separation/capsule recovery is permitted.
- 7 STAGED SEPARATION — execute only the designated release set for the specific emergency state.
- 8 CAPSULE CLEARANCE — verify physical separation and identify any attached components.
- 9 RECOVERY DEPLOYMENT — authorize parachute/recovery system within validated envelope.
- 10 DESCENT MANAGEMENT — monitor attitude, rates, canopy state and landing-support readiness.
- 11 LANDING ATTENUATION — deploy/condition energy absorbers as required.
- 12 POST-LANDING SURVIVAL — stabilize, locate, communicate, protect occupants and support rescue.
- 13 AUDIT — preserve event data, release states, sensor states and command history for engineering analysis.

Representative state-dependent behavior: an isolated engine failure should not automatically trigger capsule separation; a major fire may prioritize fuel isolation and selective separation; loss of control may require a different separation envelope; structural damage may require immediate transition; a combined catastrophic event may require the full survival chain.

16. INDEPENDENCE, FAILURE MODES AND COMMON-CAUSE CONTROL

The survival architecture introduces new hazards as well as new protections. The following failure classes shall be treated as explicit design cases.

Failure / Hazard	Required Design Response
Unintended separation	Mechanical locks, electrical inhibits, command validation, monitoring and quantitative safety objective.
Partial separation	State sensing, asymmetric-load analysis and defined degraded recovery mode.
Asymmetric separation	Dynamic analysis, CG/inertia update and post-release control assessment.
Release mechanism failure	Independent path or defined safe fallback; no assumption of perfect release.
Sensor loss / bad data	Validity monitoring, redundant/diverse sensing and safe-state logic.
Automatic detection failure	Independent backup and manual crew initiation.
Power-bus failure	Independent survival power where required for critical functions.
Communication loss	Local deterministic safety logic; communication shall not be the sole authorization path.
AI/model failure	Deterministic safety logic remains authoritative.
Fire during separation	Fuel isolation, fire containment and release sequencing.
Released component collision	Trajectory analysis, clearance envelopes and controlled sequencing.
Parachute deployment outside envelope	Hard inhibits and validated deployment conditions.
Landing-support failure	Eight-support redundancy and asymmetric/failed-leg structural cases.
Common-cause EMI/lightning/fire/structural damage	Dedicated common-cause analysis and physical/electrical segregation.

Independence is not established merely because two boxes, sensors or switches exist. The verification program shall assess common power, common wiring routes, common structure, common software, common sensor physics, common fire zones and common environmental vulnerabilities.

17. INTELLIGENT MONITORING, PREDICTION AND CONTROL BOUNDARY

LSH V4.0 may incorporate advanced health monitoring, digital-twin estimation, predictive fault detection and AI-assisted decision support. These functions shall remain bounded by deterministic safety authorization.

Recommended information/control chain:

State / Health Estimation → Emergency Prediction / Forecast → Risk State → Candidate Recovery Actions → Resilience-First Decision → Deterministic Safety Validation → Interlock / Precondition Verification → Controlled Actuation → Post-Action Monitoring → Rollback / Fallback → Immutable Event Record

- AI may detect patterns, estimate degradation and rank candidate actions.
- AI may not directly bypass interlocks or safety limits.
- AI confidence shall not be treated as safety authorization.
- Deterministic protection and safety logic remains authoritative.
- Human emergency authority remains non-bypassable.
- Model failure shall lead to a defined degraded mode.
- Critical command paths shall have software/configuration integrity controls.
- All emergency decisions shall generate an auditable event record.

18. PRINCIPAL SYSTEM INTERFACES

Interface	Definition	Control Requirement
CAP-AF	Capsule ↔ airframe structural attachment	Normal-flight load path + controlled emergency release + positive retention.
CAP-PAR	Capsule ↔ parachute system	Deployment only after validated capsule state and clearance.
CAP-LG	Capsule ↔ eight landing supports	Primary crash/landing load path with energy absorption.
JET-PWR	Independent release power	Segregated from common single-point power failure where required.
JET-CMD	Automatic/backup/manual commands	Independent paths and protected actuation.
JET-SNS	Release-state sensing	Positive confirmation of attached/released state.
FUEL-ISO	Fuel isolation ↔ separation	Isolation before or coordinated with release according to hazard analysis.
REC-LOC	Post-landing locator/communications	Independent emergency power and antenna strategy.
AV-CTRL	Avionics/control ↔ Safety Kernel	Deterministic authorization boundary.

All interfaces crossing survival boundaries shall be documented in Interface Control Documents and assessed for common-cause vulnerability. Connector, harness, hydraulic, fuel and structural interfaces shall be designed so that emergency separation does not create an uncontrolled hazard.

19. VERIFICATION, VALIDATION AND EVIDENCE PROGRAM

V4.0 deliberately separates conceptual definition from demonstrated performance. The following evidence gates are required before any high-confidence safety or performance claim is made.

Gate	Evidence Objective
G0 Architecture Freeze	Confirm safety philosophy, system boundaries, interfaces and emergency states.
G1 Analytical Closure	Mass, CG, inertia, rotor sizing, power, fuel, structural load paths and separation kinematics.
G2 Subsystem Demonstration	Release mechanisms, locks/inhibits, energy absorbers, parachute components, sensors and independent power.
G3 Integrated Ground Demonstration	End-to-end sequence without flight exposure; verify timing, state sensing and capsule clearance.
G4 Dynamic / 6-DOF Simulation	Vehicle and capsule dynamics through representative emergency states, partial/asymmetric release and parachute inflation.
G5 Structural / Crash Testing	Capsule, attachment points, seats, landing supports, energy absorbers and occupant-protection evidence.
G6 Recovery System Testing	Canopy deployment, inflation loads, failure cases, stability and landing performance.
G7 Flight Development	Controlled flight tests within approved envelope; validate normal and emergency recovery behavior.
G8 Certification / Qualification	Evidence package against the selected certification basis and applicable standards.

Terms shall be used carefully: 'specified' means defined in the design; 'analyzed' means a documented analytical result; 'simulated' means supported by a model; 'tested' means demonstrated in a defined test; 'verified' means evidence shows compliance with a requirement; 'validated' means suitability has been established for the intended use; 'qualified' and 'certified' require the applicable formal processes.

20. 6-DOF EMERGENCY DYNAMICS PROGRAM

A dedicated six-degree-of-freedom simulation shall be developed for the aircraft, separated capsule and relevant released components. The simulation shall include translation and rotation, component release impulses, aerodynamic forces, rotor/propulsor effects where relevant, parachute inflation and terrain interaction.

- Aircraft pre-release attitude, airspeed, altitude and angular-rate envelopes.
- Release impulse and mechanism timing.
- Component mass, CG and inertia before and after each release.
- Asymmetric and partial release configurations.
- Main rotor and tail rotor release cases only where under active study.
- Capsule aerodynamic coefficients across relevant attitudes.
- Parachute deployment and inflation transient loads.
- Canopy failure/degradation cases.
- Ground-contact kinematics with slope and uneven terrain.
- Landing-support deformation and energy absorption.
- Occupant acceleration/pulse proxies for early screening.
- Collision envelopes between released hardware, capsule and parachutes.

The 6-DOF model is a core verification tool, not a substitute for physical testing. Model uncertainty, sensitivity and correlation to test data shall be documented.

21. ENGINEERING / IP VALUE SURFACES

The following architecture elements are candidate IP extraction surfaces. Their legal novelty and patentability are NOT established by this document and require a professional prior-art search and claim analysis.

Candidate IP Element	Technical Content
IP-ELEMENT-0001	Occupied survival boundary integrated into a 20-ton-class rotorcraft architecture.
IP-ELEMENT-0002	State-dependent selective separation of high-mass and fire-hazard systems from an occupied survival capsule.
IP-ELEMENT-0003	Deterministic Safety Kernel authorizing separation/recovery based on physical preconditions.
IP-ELEMENT-0004	Six-layer survival architecture combining diverse initiation, physical separation, impact attenuation and post-event recovery.
IP-ELEMENT-0005	Eight-support capsule-integrated impact attenuation architecture.
IP-ELEMENT-0006	Coordinated multi-mode recovery chain: control recovery → autorotation → selective separation → capsule recovery → ground attenuation
IP-ELEMENT-0007	State-aware post-separation management for partial/asymmetric release states.
IP-ELEMENT-0008	Optional coordinated residual-airframe recovery as a separate survival mode.
IP-ELEMENT-0009	Immutable emergency event record linking detection, authorization, release states and recovery status.

22. PROPOSED TOP-LEVEL REQUIREMENT SET

ID	Requirement / Target	Status
LSH-REQ-SYS-001	Aircraft shall provide 10–12 occupant provisions within the selected cabin/capsule configuration.	[REQ / TARGET]
LSH-REQ-SYS-002	Generation 1 shall use a single main rotor and tail rotor architecture.	[REQ]
LSH-REQ-SYS-003	Design mass shall be based on a 20,000 kg target configuration.	[TARGET]
LSH-REQ-ROT-001	Main rotor diameter target shall be 21.0 m.	[TARGET]
LSH-REQ-ROT-002	Tail rotor diameter target shall be 3.5 m.	[TARGET]
LSH-REQ-RNG-001	Design range target shall be 1,000 km subject to mission closure.	[TARGET]
LSH-REQ-CAP-001	Occupied survival capsule shall constitute the final landing boundary in capsule-recovery mode.	[REQ]
LSH-REQ-CAP-002	Capsule shall include structural protection, occupant restraint, impact attenuation and post-event survival provisions.	[REQ]
LSH-REQ-LG-001	Capsule shall use eight short fixed mechanical landing supports as the preliminary architecture.	[TARGET]
LSH-REQ-JET-001	Rapid separation shall be selective and state-dependent.	[REQ]
LSH-REQ-JET-002	Automatic initiation and independent manual initiation shall be provided.	[REQ]
LSH-REQ-JET-003	Unintended separation shall be treated as a catastrophic hazard.	[REQ]
LSH-REQ-JET-004	No component shall be universally jettisonable without structural, dynamic, fire, control and collision analysis.	[REQ]
LSH-REQ-JET-005	Partial/asymmetric separation shall be explicitly modeled and detected.	[REQ]
LSH-REQ-SAF-001	Deterministic safety authorization shall be independent of AI confidence.	[REQ]
LSH-REQ-SAF-002	Human emergency authority shall not be bypassable.	[REQ]
LSH-REQ-REC-001	Capsule recovery shall use a distributed multi-canopy architecture subject to sizing.	[REQ / STUDY]
LSH-TGT-JET-001	Validated command to verified capsule-clear state target ≤ 6 s.	[TARGET / OPEN]

23. CRITICAL OPEN ITEMS — V4.0

- 1 Close the 20,000 kg mass model using component-level estimates and uncertainty.
- 2 Close the 10–12 occupant capsule geometry and bottom-up mass.
- 3 Select/size main rotor blade geometry, RPM, solidity, control system and transmission.
- 4 Close tail rotor sizing against torque and directional-control requirements.
- 5 Select candidate engines and close total installed power and OEI performance.
- 6 Develop 1,000 km mission fuel model including reserves.
- 7 Develop full 6-DOF aircraft/capsule/released-component model.
- 8 Define the exact set of jettisonable components for each emergency state.
- 9 Prove safe separation trajectories and post-jettison control for each candidate component.
- 10 Develop independent automatic and manual command architecture and common-cause analysis.
- 11 Define Safety Kernel requirements and deterministic implementation boundary.
- 12 Size the distributed parachute system and establish canopy failure tolerance.
- 13 Evaluate optional residual-airframe parachute versus added mass, drag and deployment risk.
- 14 Size the eight landing supports and energy absorbers, including asymmetric failures.
- 15 Perform capsule structural, crash, fatigue and damage-tolerance analyses.
- 16 Define occupant injury criteria and test instrumentation.
- 17 Define emergency power and post-separation electrical architecture.
- 18 Define fire, fuel isolation and thermal protection architecture.
- 19 Define emergency egress, rescue, locator and communication provisions.
- 20 Establish certification basis and verification matrix.

These open items are not weaknesses to be hidden; they are the correct engineering gates that separate a strong conceptual IP baseline from an unsupported performance claim.

24. ENGINEERING BASIS AND EXTERNAL REFERENCE NOTES

The V4.0 baseline is derived primarily from the user's prior LSH Document 00 Rev3.2 and the higher-maturity rapid-jettison Generation 1 document. The previous baseline established the six-layer survival philosophy, independent automatic/manual initiation, controlled separation, eight landing supports, capsule-as-final-landing-boundary concept, six-second development objective and the need to treat partial/asymmetric separation as first-class states.

The current V4.0 update changes the Generation 1 planning point to 20,000 kg, 21 m main rotor, 3.5 m tail rotor and 1,000 km range target. The numerical 20-ton mass accounting in this revision is a new preliminary study allocation and must be replaced by a bottom-up mass model during engineering development.

Current EASA CS-29 material was reviewed as an external engineering reference. The EASA material emphasizes that jettisonable configurations require satisfactory separation characteristics, including clear trajectory, no damaging interaction, no adverse/uncontrollable aircraft reaction, acceptable post-jettison stability/control and acceptable post-jettison performance; the V4.0 verification program therefore treats these as core analysis/test concerns rather than assuming safe release.

External reference: EASA, Easy Access Rules for Large Rotorcraft (CS-29), Amendment 12, March 2026. <https://www.easa.europa.eu/en/document-library/easy-access-rules/online-publications/easy-access-rules-large-rotorcraft-cs-29-12>

Source baseline files used: LSH_00_Lightweight_Safe_Helicopter_Gen1_Rev3_2.pdf and USCC_Document_00_Lightweight_Jettison_Gen1_Rev1.3_HigherMaturity-2.pdf. The source documents explicitly state that their numerical performance values are preliminary and that independence, jettison timing, parachute sizing, impact loads and mass properties remain verification gates.

25. V4.0 DESIGN PHILOSOPHY — CONTROLLED SURVIVAL, NOT A SINGLE DEVICE

The defining engineering proposition of LSH V4.0 is the integration of multiple survival mechanisms into one state-aware architecture. The aircraft is not intended to depend solely on a parachute, solely on autorotation, solely on a detachable cabin or solely on automatic software.

The intended chain is:

PREVENT / MONITOR → DETECT → CONFIRM → ASSESS → RECOVER IF POSSIBLE → SAFETY AUTHORIZE → SELECTIVELY SEPARATE → VERIFY CAPSULE CLEARANCE → DEPLOY RECOVERY → MANAGE DESCENT → ABSORB IMPACT → SURVIVE / LOCATE / RESCUE → AUDIT

The architecture becomes especially valuable when several failure mechanisms occur together. A single-engine failure may be handled by normal redundancy or autorotation; a fire may require fuel isolation and selective separation; a loss of control may require a different survival state; a combined fire/control/structural event may require the full capsule recovery chain.

The V4.0 baseline therefore treats safety as a system architecture and verification problem. The strongest future evidence will come from a closed mass model, rotor/propulsion analysis, 6-DOF simulation, deterministic safety logic, separation mechanism tests, capsule structural tests, parachute tests and progressively integrated demonstrations.

Document disposition: V4.0 is suitable as the controlled conceptual/master engineering baseline for the next engineering phase. It should not yet be presented as a certified aircraft specification or as proof of achieved 20-ton performance, 1,000 km range or survival probability.

END OF DOCUMENT 00 — LSH V4.0 GENERATION 1